

Schwarz Coupling for Solid Dynamics in Norma: Transmission Conditions and Interface Energy

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Abstract

This note is the reference for the Schwarz transmission conditions implemented in Norma and for the measurements of their energy behavior in dynamic solid mechanics. It describes the conditions as they are implemented now, states the requirements that an energy-consistent coupling must satisfy, and summarizes the measurements on two benchmarks: a cantilever beam released from a bent shape, with nonconforming meshes at a flat interface, and nested cylinders of three materials launched against a rigid surface, with a closed curved interface. The findings are: Dirichlet transmission on nonconforming meshes injects energy and ends in element inversion on both benchmarks; the classical Robin–Robin condition, which has no absorbing term, does the same, usually sooner, at either of two Robin parameters; the impedance condition on an overlap boundary is exact on conforming meshes but cannot be made energy consistent on nonconforming ones, because the two sides exchange data on two different surfaces through operators that are not adjoint; and the adjoint-paired impedance condition on a shared interface, which transfers a single d’Alembert interface reaction through adjoint operators built from one cross-mass matrix, never injects energy on either benchmark, is exact at mild mesh contrast, and dissipates at strong contrast at a rate set by the part of the interface velocity jump that the two trace spaces cannot represent in common. The note records the explicit–explicit and different-time-step variants of that coupling, the behavior of the relaxation and Aitken acceleration of the Schwarz iteration, and the open questions, chiefly the stalled interface jump on the cylinders and the dependence of the converged solution on the Robin parameter.

This document supersedes the July–August 2026 version of this note and the three earlier notes it consolidated. Measurements that motivated design decisions but concern superseded variants are kept only in summary.

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1 Setting

1.1 Decompositions and notation

A solid body $\Omega \subset \mathbb{R}^3$ is decomposed into two subdomains Ω_1 and Ω_2 that are integrated independently and coupled by a Schwarz iteration. The Schwarz framework itself (time controller, subdomain solves, convergence criteria) follows Mota, Tezaur, and Phlipot [1], and its contact variant Mota et al. [2]. This note concerns the transmission conditions imposed on the coupling boundaries and their energy behavior. Norma supports two geometric configurations.

Nonoverlap. The subdomains partition Ω and share an interface $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$. Let $\mathbf{n}_k$ be the outward unit normal to Ω_k on Γ (so $\mathbf{n}_1 = -\mathbf{n}_2$) and $\mathbf{t}_k = \boldsymbol{\sigma}_k \mathbf{n}_k$ the traction on the side of Ω_k . The coupled solution must satisfy

$$\mathbf{u}_1 = \mathbf{u}_2 \quad (\text{kinematic continuity}), \quad (1)$$

$$\mathbf{t}_1 + \mathbf{t}_2 = \mathbf{0} \quad (\text{traction balance}). \quad (2)$$

The Dirichlet–Neumann method imposes (1) on one side and (2) on the other. The Robin and impedance conditions of Sec. 2 impose on each side a linear combination of displacement, velocity, and traction.

Overlap. The subdomains intersect. Each Ω_k has a portion of its boundary, $\Gamma_k \subset \Omega_j$ with $j \neq k$, in the interior of the partner. That boundary is not shared with Ω_j , so no mutual traction is directly available there; the partner provides a well-defined interior state at every point of Γ_k , and the conditions of Sec. 3 are built from that state.

Throughout, $\mathbf{W}_k = \int_{\Gamma_k} \mathbf{N}_k \mathbf{N}_k^\top d\Gamma$ is the boundary mass matrix of side k , $\mathbf{P}$ a kinematic transfer operator that maps partner data to the interface nodes of a side, the subscript j denotes the partner of side k , and $[\![\mathbf{q}]\!] = \mathbf{q}_k - \mathbf{q}_j$ is the interface jump of a field $\mathbf{q}$, own side minus partner, on the coupling surface of side k . All interface operators are assembled once at initialization on the reference configuration with the facet Gauss rule of the receiving side set, and are not updated with the deformation.

1.2 Vocabulary of the multi-domain controller

The controller advances in *stops*. Within a stop each subdomain advances to the stop end in its own *substeps*, and the *Schwarz iteration* repeats a *pass* over the subdomains (one solve of each, with interface data exchanged between them) until the exchanged data converge; the stop is then committed. The exchanged datum may be relaxed between Schwarz iterations with a factor $\theta \in (0, 1]$, fixed or adapted by Aitken extrapolation (Sec. 5.4). A stop that spans several substeps is a *window* (Sec. 6.5). A mesh pair is *nested* when every coarse node coincides with a fine node. Mesh ratios are written $1:r$ with $r = h_{\text{fine}}/h_{\text{coarse}}$.

1.3 Benchmarks

Cantilever beam. A linear elastic bar ($E = 6.895$ GPa, $\nu = 0.25$, $\rho = 2768$ kg/m³; 254 mm long, 25.4 mm tall, 12.7 mm wide) is clamped at one end and released at $t = 0$ from the parabolic bending displacement $u_y = 0.3937 x^2$ (tip deflection 25.4 mm), with initial energy 1506 J. The subdomain time step is 1 μ s; horizons are 1 and 10 ms. Subdomain integrators are trapezoidal Newmark (implicit, I) and central difference (explicit, E), in the pairings II, IE, EI, and EE (first letter: clamped subdomain). Overlap decompositions split the bar lengthwise with overlap widths of 50.8, 25.4, and 12.7 mm; the nonoverlap decomposition splits it at $x = 127$ mm. The free subdomain keeps $h = 6.35$ mm and the clamped subdomain is coarsened to $6.35/r$ mm; the families used are $r = 1.0, 0.9, \dots, 0.2$ (the mesh-ratio sweep of 2026-07) and $r \in \{0.9, 0.75, 0.5\}$ (the 2026-09 rerun with the current code and the Robin–Robin study). The Schwarz iteration is converged to a relative tolerance of 10^{-8} at every stop.

Nested cylinders. A shell (outer radius 88.9 mm, height 330 mm, wall 12.7 mm; $E = 200$ GPa, $\rho = 7823$ kg/m³) contains a soft filler (radius 76.2 mm, height 305 mm; bulk modulus 5.99 GPa, shear modulus 71.7 MPa, $\rho = 1843$ kg/m³) that contains an offset core (radius 38.1 mm, height 102 mm, 19 mm above center; $E = 100$ GPa, $\rho = 4915$ kg/m³). All three are neohookean. The P-wave impedance of the filler, 3.35 MPa s/m, is one fiftieth of the shell’s. The bottom face is fixed in z and the whole assembly is launched at -35 m/s in z , so a compressive front runs up through the assembly. The decomposition splits the filler at the boundary of a cylindrical region around the core (radius 57 mm, height 229 mm): the interface has a lateral surface and two caps and lies in one material. The overlap version grows the inner region and shrinks the outer cavity by 7.6 mm each. The inner mesh is 5.1 mm and the outer 6.35 mm (ratio 1:0.8, nonconforming), 17 to 26 thousand elements per subdomain; the monolithic reference has 59 thousand elements at 5.1 mm. Every subdomain uses central difference with 0.5 μ s stops (Courant number

0.5) to a horizon of 5 ms. One Cubit journal with a refinement parameter generates all meshes (`examples/jmp/concentric-cylinders`).

Explicit stability. Central difference in Norma caps its step at $\text{CFL} \times h_{\min}/c_p$, where $h_{\min}$ is the smallest element characteristic diameter (twice the mean distance from the element centroid to its nodes) over the block and c_p the P-wave speed of the block material; a step longer than the cap is reached by substeps. On the cantilever meshes that diameter is about 1.7 times the edge length, so the cap at $\text{CFL} = 1$ exceeds the true stability limit; the cantilever explicit runs at $1 \mu\text{s}$ operate at 0.27 of the measured limit ($3.6 \mu\text{s}$ on the 6.35 mm mesh) and are unaffected by it. A Courant number of 0.5 or less makes the cap protective on hexahedral meshes, and the cylinders use 0.5.

1.4 Energy metrics

Blended energy. Summing the energies of the two subdomains counts the overlap region twice, and the double count alone produces apparent excursions of $\pm 13\%$ as a pulse crosses the overlap. All energies in this note count the overlap once through a partition of unity: the production output (`blended energy output: true, src/overlap_energy.jl`) weights each subdomain by the ratio of its distance to its own Schwarz boundary to the sum of both distances, a smooth Arlequin partition, and nonoverlap decompositions receive unit weights. Results are reported as $E(t)/E(0)$ or as the drift $E(t)/E(0) - 1$ in percent.

Staggered energy for central difference. Central difference conserves, on linear problems, the staggered lumped-mass energy

$$E_n = \frac{1}{2} \dot{\mathbf{u}}_{n-\frac{1}{2}}^T \mathbf{M}_L \dot{\mathbf{u}}_{n+\frac{1}{2}} + \text{SE}(\mathbf{u}_n) = \frac{1}{2} \dot{\mathbf{u}}_n^T \mathbf{M}_L \dot{\mathbf{u}}_n - \frac{\Delta t^2}{8} \mathbf{a}_n^T \mathbf{M}_L \mathbf{a}_n + \text{SE}(\mathbf{u}_n), \quad (3)$$

with $\mathbf{M}_L$ the lumped mass and SE the stored energy. The consistent-mass kinetic energy at integer steps misrepresents the short-wavelength content of a sharp front: a monodomain central-difference run of the cantilever reads -8.2% at 1 ms in that metric and $\pm 0.000\%$ in (3). Norma's energy output evaluates (3) for central-difference subdomains, and every explicit result below uses it. The correction term is negative, so when grid-frequency content accumulates at the stability limit ($\omega \Delta t \rightarrow 2$) the metric goes negative while displacements and stored energy stay bounded; the note calls that state *bounded and noisy*.

2 Transmission conditions on a shared interface

2.1 Robin coupling

On each side of the interface a Robin parameter $\alpha > 0$ is prescribed and the condition

$$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad k = 1, 2, \quad (4)$$

is imposed, with the datum built from the partner's previous Schwarz iterate, $\mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}$. At the fixed point $\mathbf{u}_k = \mathbf{u}_j$ and $\mathbf{t}_k = -\mathbf{t}_j$, so (4) reproduces (1) and (2). Discretely, the self term adds the boundary stiffness $\alpha \mathbf{W}$ to the tangent and the partner datum enters the boundary force as $-\mathbf{t}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$. The parameter α has units of stiffness per unit area (N/m^3) and interpolates between a Dirichlet constraint ($\alpha \rightarrow \infty$) and a Neumann condition ($\alpha \rightarrow 0$); the classical choice is $\alpha \sim E/h$.

The input keyword is **Schwarz RR nonoverlap**. It requires a positive **robin parameter**, rejects **impedance scale**, defaults to **adjoint pairing: false** (the classical per-side transfer, which allows different α on the two sides), and warns when combined with a dynamic integrator, because the condition reflects every wave (Sec. 2.3) and pumps energy at a dynamic interface (Secs. 6.7 and 7). It is intended for quasi-statics and for comparison studies. Both nonoverlap keywords are implemented by the same boundary-condition type; the Robin condition is the $Z = 0$ member of the impedance family, and the two keywords differ only in defaults and input guards.

2.2 Impedance coupling

For a semi-infinite elastic bar with density ρ and P-wave modulus $M = \lambda + 2\mu$, an outgoing wave $u(x, t) = f(x - c_p t)$ with $c_p = \sqrt{M/\rho}$ satisfies

$$\sigma + Z \dot{u} = 0, \quad Z = \rho c_p = \sqrt{\rho(\lambda + 2\mu)}, \quad (5)$$

the Lysmer–Kuhlemeyer nonreflecting condition [3]. The impedance Schwarz condition adds this absorbing term to (4):

$$\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k \quad \text{on } \Gamma, \quad \mathbf{g}_k^{(n+1)} = -\mathbf{t}_j^{(n)} + Z \dot{\mathbf{u}}_j^{(n)} + \alpha \mathbf{u}_j^{(n)}. \quad (6)$$

At the fixed point the velocity and displacement terms cancel and (6) again reproduces continuity and traction balance. The impedance is computed from the material of the element block that owns the interface side set, $Z_k = \sqrt{\rho_k(\lambda_k + 2\mu_k)}$, multiplied by the input **impedance scale** (default 1, required positive); α is an optional input (default 0). The keyword is **Schwarz impedance nonoverlap**, with **adjoint pairing: true** as its default (Sec. 5).

Under Newmark integration the end-of-step velocity is linear in the end-of-step displacement with coefficient $c_v = \gamma/(\beta \Delta t)$, so the self term contributes the tangent $(Z c_v + \alpha) \mathbf{W}$ and the self force $\mathbf{W}(Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$. In the quasi-static limit $c_v = 0$ and the condition reduces to (4); a quasi-static run with $\alpha = 0$ is a pure Neumann condition and does not couple the subdomains kinematically.

2.3 Reflection at the interface

Place the interface at $x = 0$ with Ω_k in $x < 0$ and consider a harmonic incident wave and its reflection. The Robin condition $\sigma + \alpha u = 0$ gives the reflection coefficient

$$R_{\text{RR}}(\omega) = \frac{i\omega Z - \alpha}{i\omega Z + \alpha}, \quad |R_{\text{RR}}(\omega)| \equiv 1, \quad (7)$$

so for any real $\alpha > 0$ the interface reflects the incoming wave completely, with a frequency-dependent phase shift. The impedance condition $\sigma + Z \dot{u} + \alpha u = 0$ gives

$$R_{\text{imp}}(\omega) = -\frac{\alpha}{2i\omega Z + \alpha}, \quad |R_{\text{imp}}(\omega)| = \frac{\alpha}{\sqrt{\alpha^2 + 4\omega^2 Z^2}} < 1, \quad (8)$$

with $R_{\text{imp}} \equiv 0$ at $\alpha = 0$. The dashpot is the only mechanism in either condition that removes energy from an interface mode; the Robin spring stores it and returns it.

3 Transmission conditions on an overlap boundary

3.1 Dirichlet coupling

The classical overlap condition imposes the partner's kinematic trace on Γ_k ,

$$\mathbf{u}_k = \mathbf{u}_j, \quad \dot{\mathbf{u}}_k = \dot{\mathbf{u}}_j, \quad \ddot{\mathbf{u}}_k = \ddot{\mathbf{u}}_j \quad \text{on } \Gamma_k, \quad (9)$$

either by pointwise interpolation (locate the partner element that contains each node in the reference configuration and evaluate the partner shape functions there) or by the L^2 projection $\mathbf{P}_{kj} = \mathbf{W}_k^{-1} \mathbf{L}_{kj}$ with $\mathbf{L}_{kj} = \int_{\Gamma_k} \mathbf{N}_k (\mathbf{N}_j^{\text{vol}})^\top d\Gamma$. Both are implemented by `SolidMechanicsOverlapSchwarzBoundaryCondition` (keyword `Schwarz overlap`). The constrained degrees of freedom add no interface tangent.

3.2 Impedance coupling

The impedance overlap condition leaves the overlap degrees of freedom free and imposes

$$\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z} (\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0} \quad \text{on } \Gamma_k, \quad (10)$$

where $\mathbf{t}_j = \boldsymbol{\sigma}_j \mathbf{n}_k$ is the partner's traction on Γ_k with the outward normal of Ω_k (the partner has no boundary there, so the converged state satisfies $\mathbf{t}_k = \mathbf{t}_j$). The own traction $\mathbf{t}_k$ is never computed: it is the natural boundary term of the weak form of Ω_k , and the condition is imposed by adding the partner datum $\mathbf{W}(\mathbf{t}_j + \mathbf{Z} \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$ as a Neumann load and the self terms $\mathbf{W}(\mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$ as an interface force on the same residual. The impedance is the P/S-split tensor

$$\mathbf{Z} = Z_p \mathbf{n}_k \otimes \mathbf{n}_k + Z_s (\mathbf{I} - \mathbf{n}_k \otimes \mathbf{n}_k), \quad Z_p = \sqrt{\rho(\lambda + 2\mu)}, \quad Z_s = \sqrt{\rho\mu}, \quad (11)$$

with ρ, λ, μ from the partner's material in the block named by `source block` (each side carries the impedance of its neighbor, the optimized-Schwarz principle that the transmission operator should approximate the neighbor's Dirichlet-to-Neumann map). The keyword is `Schwarz impedance overlap`.

The partner traction. The monodomain solution has continuous kinematics and a nonzero traction on the artificial surface Γ_k , and satisfies (10) identically. Without the term $-\mathbf{t}_j$ the condition becomes a relative dashpot that the monodomain solution fails by exactly that traction, and the converged coupled solution then carries a spurious velocity jump of about $\mathbf{t}/Z$ whose dissipated power $|\mathbf{t}|^2/Z$ acts as permanent interface damping (59% loss over 1 ms on the conforming cantilever; Sec. 6.1). Because Γ_k is an interior surface of the partner mesh, the partner's assembled internal force cannot supply $\mathbf{t}_j$ there. Norma evaluates it in one of two ways (`partner traction`, default `auto`). On node-aligned interfaces the weak traction is the consistent nodal reaction of the single layer of partner elements exterior to Γ_k ,

$$[\mathbf{W} \mathbf{t}_j]_i = - \sum_{e \in \mathcal{E}_i^{\text{ext}}} (\mathbf{f}_e^{\text{int}} + \mathbf{M}_e \ddot{\mathbf{u}}_e)_i, \quad (12)$$

which the monodomain solution satisfies identically, mesh-scale content included (implemented by `build_consistent_traction_patch!`). Otherwise the partner's consistently recovered nodal stress is transferred to Γ_k and contracted with the outward normal; recovery is enabled automatically for models that carry this condition.

Transfer and assembly. The kinematic fields and the recovered stress are transferred by the L^2 projection by default (**transfer: variational**; pointwise interpolation is a legacy option, and the two coincide on node-aligned interfaces). On non-aligned interfaces the whole partner datum is assembled on the partner's offset surface from the partner's own nodal values and transferred with one surface-to-surface operator (**characteristic_partner_terms**). The self tangent consists of the nodal blocks $W_{ij} (c_v \mathbf{Z}(\mathbf{n}_j) + \alpha \mathbf{I})$. A scalar or step-dependent multiplier on $\mathbf{Z}$ (**impedance scale**) remains available; with the traction term in place the physical scale 1 is the conserving one, and the historical use of the scale as a per-problem parameter, which compensated the missing traction, is obsolete. With **impedance scale: 0** the condition is the Robin–Robin overlap condition $\mathbf{t}_k - \mathbf{t}_j + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0}$; no separate keyword exists for it.

3.3 Summary of the four conditions

Table 1: Transmission conditions on a shared (nonoverlap) interface.

	Robin	Impedance
Boundary condition	$\mathbf{t}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$	$\mathbf{t}_k + Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k = \mathbf{g}_k$
Self tangent	$\alpha \mathbf{W}$	$(Z c_v + \alpha) \mathbf{W}$
Self force	$\alpha \mathbf{W} \mathbf{u}_k$	$\mathbf{W} (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$
Partner datum	$-\mathbf{t}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$	$-\mathbf{t}_j + Z \mathbf{W} \mathbf{P} \dot{\mathbf{u}}_j + \alpha \mathbf{W} \mathbf{P} \mathbf{u}_j$
Z	(none)	interface block material; harmonic mean per pair
α	input, required, per side	input, optional, one value per interface
Transfer	per side (legacy)	adjoint pair from one cross-mass matrix
Reflection $ R $	$\equiv 1$	< 1 ; 0 at $\alpha = 0$
Keyword	Schwarz RR nonoverlap	Schwarz impedance nonoverlap

Table 2: Transmission conditions on an overlap boundary.

	Dirichlet	Impedance
Boundary condition	$\mathbf{u}_k = \mathbf{u}_j$ (pointwise or L^2)	$\mathbf{t}_k - \mathbf{t}_j + \mathbf{Z} (\dot{\mathbf{u}}_k - \dot{\mathbf{u}}_j) + \alpha (\mathbf{u}_k - \mathbf{u}_j) = \mathbf{0}$
Self tangent	(DOFs constrained)	$W_{ij} (c_v \mathbf{Z}(\mathbf{n}_j) + \alpha \mathbf{I})$
Self force	(none)	$\mathbf{W} (\mathbf{Z} \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$
Partner datum	$\mathbf{u}_k \leftarrow \mathbf{P} \mathbf{u}_j$	$\mathbf{W} (\mathbf{t}_j + \mathbf{Z} \dot{\mathbf{u}}_j + \alpha \mathbf{u}_j)$
Partner traction	(none)	consistent (12) or recovered stress
$\mathbf{Z}$	(none)	partner material, P/S split
α	(none)	input, optional
Keyword	Schwarz overlap	Schwarz impedance overlap

4 Requirements for energy consistency

Three results from the literature fix the structure of an energy-consistent coupling; the measurements of Secs. 6–7 confirm each of them.

The interface work identity. For Newmark-family subdomains coupled by interface tractions, the discrete energy balance over a step contains the interface term (Gravouil–Combescure [4, Eq. (34)]; Karimi–Nakshatrala [6, Eq. (84)])

$$E_{\text{int}} = \sum_{\text{sides } k} \frac{1}{\Delta t_k} \llbracket \dot{\mathbf{u}}_k \rrbracket^T \mathbf{C}_k^T \llbracket \Lambda_k \rrbracket, \quad (13)$$

the work of the transmitted tractions through the kinematic jump, each side on its own grid ($\mathbf{C}_k$ is the interface restriction of side k , Λ_k the traction impulse transmitted to it, and $\llbracket \cdot \rrbracket$ here the change across the step). The term vanishes if and only if the traction is one field acting equal and opposite on the two sides through adjoint trace operators, with velocity continuity enforced at instants shared by both time grids. When the pairing is broken the term has no sign.

Representability on nonconforming grids. Gander, Halpern, and Nataf [7] prove convergence of Schwarz waveform relaxation with impedance transmission on nonmatching grids under three conditions: Robin-type (not Dirichlet) transmission, a consistent boundary discretization of the transmission operator, and an intergrid transfer that is an L^2 contraction (a projection qualifies; pointwise interpolation does not). The choice $Z = \rho c$ is the exact transparent operator in one dimension. On nonconforming grids the coupled system defines the converged solution; there is no monodomain scheme it equals. Continuity holds only through the projection, and the component of the fine trace orthogonal to the coarse trace space is never matched: the dashpot then absorbs that persistent component as an absorbing boundary would, which is sign-definite dissipation, required for stability. Achdou, Japhet, Maday, and Nataf [8] locate the remedy in a common trace space for the datum, not in a better Z .

The adjoint requirement. Energy-stable discontinuous Galerkin couplings [9, 10, 11] compute one interface state on a common description of the interface and use it in both weak forms; the interface power then reduces to

$$P = - \int_{\Gamma} \llbracket \dot{\mathbf{u}} \rrbracket^T \mathbf{Z} \llbracket \dot{\mathbf{u}} \rrbracket dS \leq 0, \quad (14)$$

which is confined to the jump and vanishes under refinement and at convergence. The discrete requirement is that the two kinematic transfers be L^2 -adjoint through one cross-mass matrix,

$$\mathbf{W}_1 \Pi_{2 \rightarrow 1} = (\mathbf{W}_2 \Pi_{1 \rightarrow 2})^T = \mathbf{B}, \quad B_{mn} = \int_{\Gamma} \varphi_m^1 \varphi_n^2 dS. \quad (15)$$

Two independent interpolations violate (15), the cross terms of the interface power do not cancel, and its sign is undetermined. Liu et al. [11] add that interface unknowns may need implicit treatment even between explicit interiors.

Consequence for the two decompositions. Condition (15) presumes one shared surface on which the two works cancel. The overlap decomposition has none: data are exchanged on two distinct surfaces Γ_1 and Γ_2 , so no pairing of the two works exists, and energy consistency could only come from agreement of the two discrete solutions in the overlap, which the differing discrete dispersion relations of the two grids preclude at fixed h (Sec. 6.3). The nonoverlap decomposition has a shared interface, and the adjoint-paired coupling of Sec. 5 implements (15) on it.

5 The adjoint-paired nonoverlap impedance coupling

This section describes the coupling as implemented (the default of `Schwarz impedance nonoverlap`; `adjoint pairing: false` restores the legacy per-side transfer). Its measured behavior is in Secs. 6.4 and 7.

5.1 Construction

Transfer operators. The cross-mass matrix $\mathbf{B}$ of (15) is integrated over the facets of the finer side (the side with more interface nodes): each facet quadrature point is mapped to the partner surface by closest-point projection in the reference configuration and the partner facet shape functions are evaluated there. This rule is exact when the coarser trace mesh is nested in the finer facets and carries facet-quadrature error where a coarse basis function kinks inside a fine facet; the partition-of-unity error $\max_i |(\Pi_k \mathbf{1} - \mathbf{1})_i|$ is logged at setup as a certificate (zero on conforming and nested interfaces). The kinematic transfers are $\Pi_1 = \mathbf{W}_1^{-1} \mathbf{B}$ and $\Pi_2 = \mathbf{W}_2^{-1} \mathbf{B}^T$, the force transfers their partner adjoints $N_1 = \Pi_2^T$ and $N_2 = \Pi_1^T$, all scalar and applied to each Cartesian component.

Exchanged traction. The weak partner reaction of side j is the restriction to its interface degrees of freedom of its assembled d'Alembert residual,

$$\lambda_j = -(\mathbf{M}_j \ddot{\mathbf{u}}_j + \mathbf{f}_j^{\text{int}} - \mathbf{f}_j^{\text{body}})|_{\Gamma}, \quad (16)$$

with $\mathbf{M}_j$ the mass matrix its integrator uses (consistent for Newmark, lumped for central difference). With this reaction the two interface rows sum exactly to the monodomain row, so the monodomain trajectory is the fixed point. The legacy coupling transferred the static reaction $-\mathbf{f}^{\text{int}}$; in dynamics the two sides' static reactions differ by the inertia of the interface nodes, each side carrying its own tributary mass, and the fixed point then differs from the monodomain discretization by a steady, step-independent interface energy loss (-9.6% per millisecond on the conforming cantilever, Table 4).

Pair impedance and Robin parameter. The pair shares one impedance, the harmonic mean $2Z_1 Z_2 / (Z_1 + Z_2)$ of the per-side material impedances $Z_k = \rho_k c_{p,k}$ of the blocks that own the two side sets, each multiplied by its side's `impedance scale`; this is the Riemann-solver weighting of [10, 11]. The Robin parameter α is one value per interface: the spring stores and returns energy conservatively only when both interface forces pair through the same coefficient, so mismatched values abort with a message naming both side sets. The pair impedance is scalar (P-wave); a tensor pair impedance has not been needed on the benchmarks studied.

Partner datum. Side k receives

$$\mathbf{g}_k = N_k \lambda_j + Z \mathbf{B}_k \dot{\mathbf{u}}_j + \alpha \mathbf{B}_k \mathbf{u}_j, \quad \mathbf{B}_1 = \mathbf{B}, \quad \mathbf{B}_2 = \mathbf{B}^T, \quad (17)$$

with the partner fields evaluated at side k 's current substep time (Sec. 5.3). The self terms, the tangent $(Zc_v + \alpha) \mathbf{W}_k$ and the force $\mathbf{W}_k (Z \dot{\mathbf{u}}_k + \alpha \mathbf{u}_k)$, are assembled with the internal force at every residual evaluation.

What the parameters change. At the fixed point the two conditions add to traction balance and subtract to $Z \dot{\delta} + \alpha \delta = \mathbf{0}$ for the interface jump δ . When the jump can close, as it does on conforming and nested meshes, the converged solution is independent of $Z > 0$ and $\alpha \geq 0$, and both act only on the convergence rate of the iteration. On nonconforming meshes the jump closes only in the projected sense of (15) and the accepted iterate has a residual jump, so the α term is a spring force on that residual, and its influence on the converged coupled solution has not been measured for the paired coupling (Sec. 8.3). The classical Robin–Robin runs of Sec. 6.7 show that with per-side transfer the converged solution does depend on α .

5.2 Explicit subdomains

With a central-difference subdomain on either side, the coupling is strongly unstable if the interface terms are evaluated at lagged time levels: the exchanged reaction (16) couples the partner’s acceleration into the interface force, an algebraic loop that the implicit pairs resolve within the Newton iteration. Following Liu et al. [11], the explicit update is closed implicitly on the interface rows only,

$$(\mathbf{M} + \gamma \Delta t Z \mathbf{W})|_{\Gamma} \mathbf{a}_{n+1} = \mathbf{M} \mathbf{a}^{\text{expl}}|_{\Gamma}, \quad (18)$$

a small symmetric positive definite solve per interface, factored once and reused for the three components while the operator is unchanged (`imex_interface_acceleration!`); the interior stays explicit. Here $\mathbf{M}|_{\Gamma}$ is the lumped mass on the interface rows, $\mathbf{W}$ the side’s own boundary mass matrix, $\gamma = \frac{1}{2}$, and $\mathbf{a}^{\text{expl}}$ the plain explicit update, whose right-hand side contains the dashpot at the predictor velocity and the Robin spring at the known end-of-step displacement; the solve replaces the predictor-velocity dashpot by the end-of-step one. The Robin spring stays explicit, so a large α can violate the central-difference stability limit on the interface rows. Interface degrees of freedom held by Dirichlet conditions keep their prescribed acceleration and enter the free rows as data.

5.3 Different time steps per subdomain

Each subdomain records full-field snapshots (displacement, velocity, acceleration, assembled internal force) at every substep of every Schwarz iteration, anchored by a snapshot of the stop-start state. When a coupling condition is evaluated at time t , each field is interpolated piecewise linearly to t (queries beyond the history clamp to its endpoints), the partner state is temporarily replaced by the interpolants, the coupling terms are evaluated, so the reaction (16) is formed from the interpolated acceleration and internal force, and the partner state is restored. This is the interpolated-continuity structure under which Gravouil and Combescure [4] prove the interface work of the subcycled exchange nonpositive and of order Δt . Two earlier defects (an off-by-one in the segment selection and a missing stop-start anchor) had reduced the exchange to end-sampled constant data; both are fixed and the measurements below use the interpolated exchange.

Three constructions that would make the coarse side’s datum exactly work-conjugate to the fine side’s interpolation, as in Prakash and Hjelmstad [5], were implemented and all fail in a partitioned exchange: the trapezoidal window average lags the interface motion by half a window and adds dissipation; the recursion $\hat{\lambda}_{n+1} = 2\bar{\lambda} - \hat{\lambda}_n$ doubles the contraction factor of the Schwarz iteration and diverges within tens of stops; and the least-squares endpoint fit is an extrapolation whose gain exceeds one outside the linear band and injects energy without bound over hundreds of stops. Only causal transfer operators are robust in a fixed-point exchange; exact cancellation of the subcycled interface work requires a direct interface solve, as Prakash and Hjelmstad use. The window-average and endpoint-fit utilities remain in `interpolation.jl` as tested building blocks.

5.4 Relaxation and Aitken acceleration

The relaxed datum is the assembled $\mathbf{g}_k$ of (17) on one side of each pair (the later-listed subdomain); a fresh relaxation slot starts from zero rather than from the incoming datum. Two Aitken forms are available (`relaxation: aitken recursive` and `aitken secant`), with `relaxation parameter` the fixed or initial factor and `aitken NO parameter` the number of initial Schwarz iterations at the input factor.

State keyed by substep slot. The relaxation state (the stored datum and the Aitken residuals) is keyed per pair and per substep time slot within the stop, and cleared at every stop (`relaxation_slot!`). Before this, one vector per pair was updated at every application of the condition, so a stop spanning several substeps blended iterates across time, a causal low-pass filter on the exchanged traction that shifted the fixed point (measured on the conforming cantilever at 10 ms: +11% to −84% with Aitken and −7% to −88% with fixed $\theta = 0.5$ as the window grew from 2 to 10 substeps). With the slot-keyed state, windowed stops reproduce the same-step trajectory to Schwarz truncation (Sec. 6.5).

Guards. On the Dirichlet-transfer paths (overlap and Dirichlet–Neumann) a fresh slot seeds the stored iterate with the incoming datum, so the first residual is exactly zero and the secant factor evaluates to −0.0; a zero factor freezes the interface, and in a chain of three or more subdomains the whole Schwarz iteration then reports convergence after one iteration with an answer wrong by 100%. The secant update is now taken only when the residual difference exceeds a floor, a factor of magnitude below 10^{-12} is recorded on the controller, and the convergence criterion refuses to accept that iteration. The recursive form no longer runs the first NO iterations unrelaxed regardless of the input factor.

Where Aitken applies. Aitken acceleration is applied only to single-slot stops (`aitken_applies`); in windowed stops the residuals of different slots belong to different times and the extrapolated factor overshoots. On the implicit cantilever pairs it reduces the Schwarz iterations by about a factor of ten against a fixed factor of 0.5 (599 against 56 recursive or 58 secant over ten stops). On the explicit cylinders pair (Sec. 7) every Aitken form diverges: the displacement residual between successive iterations collapses while the interface velocity jump stalls at a finite value, the extrapolated factor exceeds 2 by the second or third iteration, oscillates between −10 and +3, and the iteration reaches the cap of 256 within the first 25 stops, after which an element inverts; the recursive form with NO = 2 and the secant form fail the same way, six stops later. A fixed factor of 0.5 converges there in 2 to 3 iterations per stop, and $\theta = 1$ (no relaxation) diverges after 0.3 ms.

6 Measurements on the cantilever beam

6.1 Overlap, conforming meshes: the partner traction

On the conforming node-aligned overlap pair (II, $\alpha = 0$, physical impedance) the energy error of the impedance overlap coupling over 1 ms decreases through four corrections, none with a parameter:

$$\underbrace{59\%}_{\text{no partner traction}} \longrightarrow \underbrace{14\%}_{\text{traction from lumped recovery}} \longrightarrow \underbrace{1.5\%}_{\text{consistent recovery}} \longrightarrow \underbrace{0.08\%}_{\text{consistent traction (12)}}$$

against the 0.1% of converged conforming Dirichlet coupling. The residual of the recovered-stress step was diagnosed with an interface-work instrumentation that separates the partner-

traction, dashpot, and Robin channels (`report_impedance_interface_work`, environment-gated, CSV through `NORMA_IMPEDANCE_WORK_CSV`): it is unchanged under a Schwarz tolerance of 10^{-11} and under halving the time step, and it halves when h is halved. The recovery error on mesh-scale content is converted by the transmission condition into a velocity jump of about $\delta t/Z$ that the dashpot dissipates at rate $|\delta t|^2/Z$ over the whole run; the consistent traction removes it (integrated dashpot work $-42.6\text{ J} \rightarrow +0.1\text{ J}$). The P/S tensor impedance (11), which appeared worse than a scalar one with recovered stress (4.1% against 1.5%), is exact with the consistent traction (0.076%): its smaller shear impedance amplified the tangential recovery error. Historical per-problem tuning of the impedance scale, whose best value on the nonconforming pair was 1.12, compensated the missing traction against transfer injection and is obsolete.

6.2 Overlap, nonconforming meshes: sign-indefinite interface work

Table 3 gives the drift at 1 ms of the impedance overlap coupling with variational transfer across nine mesh ratios, three overlap widths, and four integrator pairings (108 runs).

Table 3: Energy drift at 1 ms of the impedance overlap coupling (variational transfer, blended energy), in percent.

ratio	overlap 50.8 mm				overlap 25.4 mm				overlap 12.7 mm			
	II	IE	EI	EE	II	IE	EI	EE	II	IE	EI	EE
1:1.0	-0.1	-8.9	-0.2	+8.7	-0.1	+5.0	-0.4	+10.2	-0.0	-15.7	-6.1	+3.6
1:0.9	+7.6	-11.7	-5.5	+8.5	-4.4	-8.1	+4.4	+9.1	+0.1	-14.8	-1.0	+4.3
1:0.8	+29.8	-15.3	+5.5	-9.5	+20.4	+45.1	-2.4	-1.9	-3.1	-20.3	-5.4	+4.7
1:0.7	+19.2	-12.8	+2.7	-11.6	+11.8	-18.6	+0.7	-5.4	+11.1	+7.3	-9.8	-17.9
1:0.6	-61.1	-54.0	-57.7	-40.4	+2.1	-18.6	-63.9	-53.4	-4.5	-34.2	-57.1	-52.7
1:0.5	-50.6	-32.3	-36.4	-56.8	-11.0	-17.6	-55.7	-54.3	-29.2	-54.0	-56.8	-52.0
1:0.4	-12.4	+3.9	-43.1	-53.1	-52.7	-50.2	-57.0	-63.8	-36.6	-52.4	-78.2	-76.2
1:0.3	-65.0	-44.5	-78.3	-57.4	+173.7	+274.7	-81.2	-78.8	-30.5	-58.6	-79.5	-79.4
1:0.2	-79.1	-76.8	-80.8	-78.9	-66.5	-66.2	-86.3	-87.4	-22.2	-45.8	-91.2	-90.3

Conforming implicit runs conserve to 0.1% at every width. Every nonconforming configuration either loses tens of percent or gains: the II column injects at mild coarsening at nearly every width (up to +29.8% at 1:0.8), the 25.4 mm IE run at 1:0.3 reaches +275%, and the drift is not monotone in the ratio (at 25.4 mm the II column moves $-52.7 \rightarrow +173.7 \rightarrow -66.5\%$ across 1:0.4, 1:0.3, 1:0.2). The sign of the interface work is decided by the mesh pair and the overlap geometry together, and no a priori bound on its sign or magnitude exists. The channel instrumentation shows the mechanism: on the 50.8 mm/1:0.5 II case with pointwise transfer the dashpot does +554 J of positive work over 1 ms, which a dissipator cannot do, so the discrete dashpot term is not a quadratic form; the two sides interpolate partner fields with unrelated operators and their dashpot terms do not pair.

Dirichlet control. The same nine mesh configurations with Dirichlet overlap coupling (II) conserve to $\pm 0.05\%$ on conforming meshes and are severely unstable on every nonconforming one (+22% to +7765% at 1 ms, most runs ending in Newton divergence before 1 ms), with pointwise and with weak transfer alike. The dashpot of the impedance condition is therefore what stabilizes a coupling that kinematic matching destabilizes, and its dissipation is the energy of the inter-grid discrepancy it absorbs.

Ten milliseconds. Of the 108 overlap runs rerun to 10 ms, 74 ended by element inversion after exponential energy growth (typically 30 to 100 times $E(0)$ at failure, between 2 and 9 ms) and most of the remainder decayed toward zero energy. Configurations that appeared dissipative at 1 ms reversed and grew. Conformity does not stabilize the mixed pairings: the conforming 25.4 mm IE run grew exponentially and failed at 6.25 ms. The only overlap configurations stable and accurate at 10 ms are the conforming II runs (+0.6, +0.6, +0.2% at the three widths; Fig. 1).

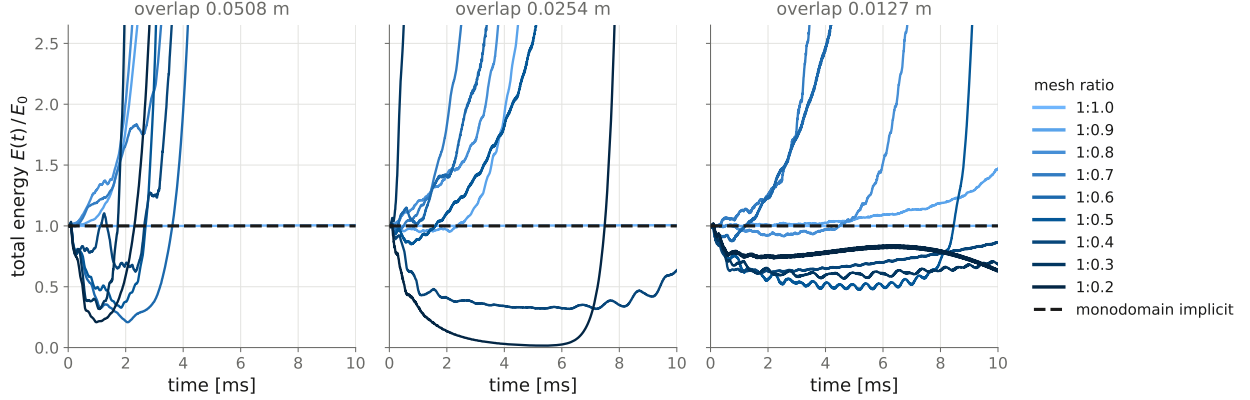


Figure 1: Overlap impedance coupling (variational transfer), implicit–implicit, 10 ms. Nearly every nonconforming configuration grows exponentially until element inversion (curves leaving the top of the axes) or decays toward zero energy; only the conforming runs follow the monodomain reference.

6.3 Overlap: remedies tried on the interface

Every remedy applied to the overlap exchange was measured on the instrumented 1 ms benchmark; none is a structural remedy, and the sequence explains why.

- **Transfer fidelity.** Replacing pointwise interpolation by the variational projection, refining the transfer quadrature, taking the consistent traction across the interface from the partner’s offset surface, and transferring the whole partner datum with one operator each *increased* the spurious dashpot work on the 1:0.75 pair (−607, −726, −857, −1129, −1084 J) while the net drift moved from −0.75% to +9.8%: pointwise interpolation and nodal stress recovery had been smoothing the transferred data, and the small net of the original scheme was a cancellation of two large channels. Over the 24 nonconforming cases of the three-ratio family the variational transfer improved 17 and reduced the mean absolute drift from 33 to 25%; it is the default because it is the contraction the convergence theory requires, not because it conserves.
- **Content-aware absorption** (content aware absorption): an additional dissipative term $\mathbf{W} \mathbf{Z} (\mathbf{I} - \hat{\Pi}) \dot{\mathbf{u}}_k$, with $\hat{\Pi}$ the $\mathbf{W}$ -orthogonal projection onto the transferable subspace, is verified (annihilates constants, vanishes on node-aligned interfaces, provably dissipative; test 106) and has no effect: it dissipates 4.6 J while every channel stays identical. The recirculating content lies inside the transferable subspace: both grids represent it, and the discrepancy comes from their different discrete dispersion relations, which no transfer construction can remove.
- **Representable dashpot** (representable dashpot): restricting the dashpot to $\mathbf{Z} \hat{\Pi} [\dot{\mathbf{u}}]$

leaves the dashpot work unchanged where the projection is nontrivial ($-1066 \rightarrow -1065$ J) and worsens the drift where it acts ($+12.5 \rightarrow +45.6\%$). Retained as a diagnostic, default off.

- **Numerical damping.** Newmark $\gamma = 0.6$ suppresses the recirculation but removes 87% of the physical energy. HHT- α (HHT **alpha**, $\bar{\alpha} \in [0, 1/3]$) at $\bar{\alpha} = 0.02$ to 0.05 converts the sign-indefinite drift into monotone decay with a loss proportional to the energy in the grid-frequency band; on this benchmark, whose release has content at every scale, the loss is 5 to 11% at 1 ms.

6.4 Nonoverlap: legacy and paired coupling

Table 4: Energy drift at 1 ms of the nonoverlap impedance coupling on the cantilever, in percent. Left: legacy per-side transfer against the adjoint-paired coupling, implicit-implicit. Right: the paired coupling in all four integrator pairings, staggered metric for explicit members.

ratio	legacy II	paired II	paired II	IE	EI	EE
1:1.0	−9.61	− 0.03	−0.03	−0.11	+0.20	−0.27
1:0.9	−6.84	− 0.03	−0.03	−0.10	+0.29	−0.27
1:0.8	−7.96	−3.59	−3.6	−4.4	−3.5	−7.3
1:0.7	−2.06	−6.55	−6.6	−6.2	−5.3	−9.8
1:0.6	−9.39	−9.71	−9.7	−9.8	−9.8	−11.8
1:0.5	−13.25	−13.32	−13.3	−13.8	−10.2	−10.5
1:0.4	−45.78	−15.13	−15.1	−17.8	−11.4	−13.3
1:0.3	+18.95	−15.08	−15.1	−11.5	−26.4	−20.2
1:0.2	+77.49	−13.26	−13.3	−20.1	−27.8	−21.4

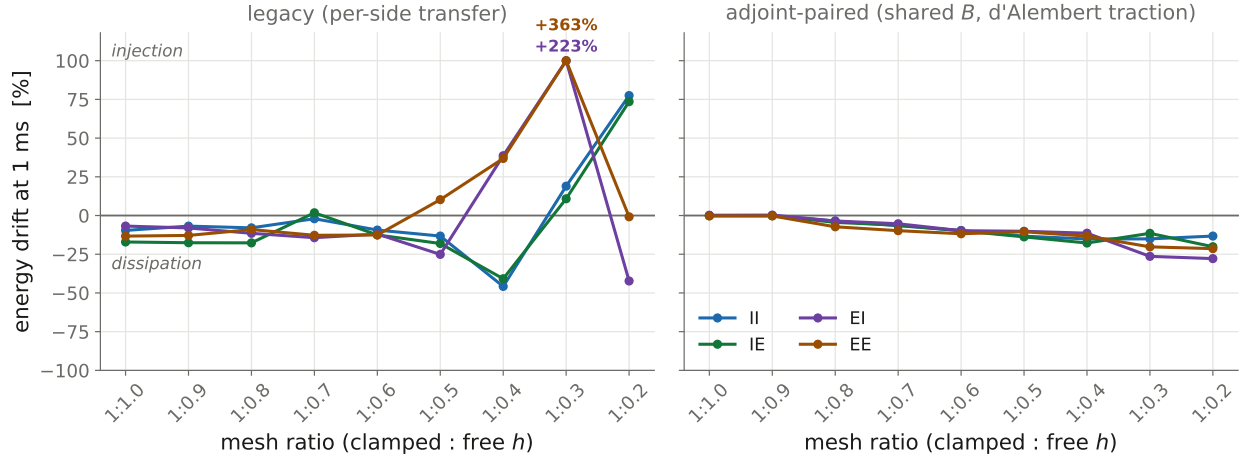


Figure 2: Nonoverlap impedance coupling, drift at 1 ms across the mesh-ratio family and the four integrator pairings. Left: legacy per-side transfer (consistent-mass metric), with injection up to +363% (EE) and +223% (EI) at 1:0.3. Right: adjoint-paired coupling (staggered metric for explicit members), sign-definite dissipation at every ratio and conservation to 0.3% on conforming and 1:0.9 meshes.

The legacy coupling loses 2 to 18% at mild contrast in every pairing, including on conforming

meshes (-9.6% II at 1:1, the static-reaction defect of Sec. 5.1, unchanged under $\alpha \rightarrow 0$, equal α , and $\Delta t/2$), and injects strongly at strong contrast. The paired coupling (Table 4, Fig. 2) never injects at any ratio in any pairing, conserves to 0.3% on conforming and 1:0.9 meshes in all four pairings, and against the monodomain explicit reference the conforming EE trajectory agrees to 0.35% pointwise over the millisecond. The residual dissipation at strong contrast is the interface jump the two trace spaces cannot represent in common, absorbed at rate $Z \int_{\Gamma} [\![\mathbf{u}]\!]^2$: at the nested ratio 1:0.5, where the legacy transfer is coincidentally adjoint and the quadrature exact, paired and legacy agree to 0.1% . The small positive EI entries are bounded transient storage in the interface spring.

Ten milliseconds. Table 5 and Fig. 3 give the 10 ms behavior. With an implicit member the conforming and 1:0.9 columns stay within 0.0 to -0.9% over ten thousand stops. At strong contrast the dissipation that read 10 to 28% at 1 ms reaches 40 to 85%: bounded and monotone, but far too large for wave-transit accuracy, since absorption of unrepresentable content proceeds at a persistent rate. The non-nested mild ratios 1:0.7 and 1:0.8 form a failure window not detectable at 1 ms: interface-local high-frequency displacement content accumulates until an interface element inverts (implicit pairs, about 5 ms, with the global energy still smooth) or central difference becomes unstable (explicit members, 2.6 to 4 ms, at 0.27 of the stability limit, so driven by the accumulated content). Reruns with fixed $\theta = 0.5$ fail identically, so the accumulation belongs to the transfer operators, whose partition-of-unity certificate is worst in this window; the nested and the strong ratios do not show it.

Table 5: Energy drift at 10 ms of the adjoint-paired nonoverlap coupling on the cantilever, staggered metric, in percent. $\dagger t$: element inversion at t ms. *: bounded and noisy (Sec. 1.4).

ratio	II	IE	EI	EE
1:1.0	+0.00	-0.9	-0.2	*
1:0.9	+0.00	-0.5	+0.1	*
1:0.8	$\dagger 5.2$	$\dagger 3.9$	$\dagger 4.0$	$\dagger 2.6$
1:0.7	$\dagger 4.9$	$\dagger 3.7$	$\dagger 3.9$	$\dagger 2.8$
1:0.6	-47.3	-56.4	-42.9	*
1:0.5	-56.8	-72.0	-40.8	*
1:0.4	-54.9	-73.9	-40.4	*
1:0.3	-65.1	-66.0	-81.1	*
1:0.2	-53.3	-73.1	-84.1	*

Explicit–explicit pairs. The EE column ends bounded and noisy at every ratio, conforming included: the stored energy stays physical (the conforming run keeps $E/E(0) \geq 0.97$ to 5 ms) while the staggered kinetic term goes negative, because central difference with $\gamma = \frac{1}{2}$ removes no grid-frequency content at any stable step and the exchange keeps introducing it. Subcycling the explicit member (free side at $\Delta t/4$) removes the accumulation: that run ends at -10.8% , the dissipation of the interpolated exchange. Subcycling has the opposite effect on implicit pairs, where the same-step conforming II run conserves to 0.00% and the subcycled one loses -20.2% over 10^4 stops (-3.6% at 1 ms), the interpolation dissipation of Sec. 5.3 accumulated over the run.

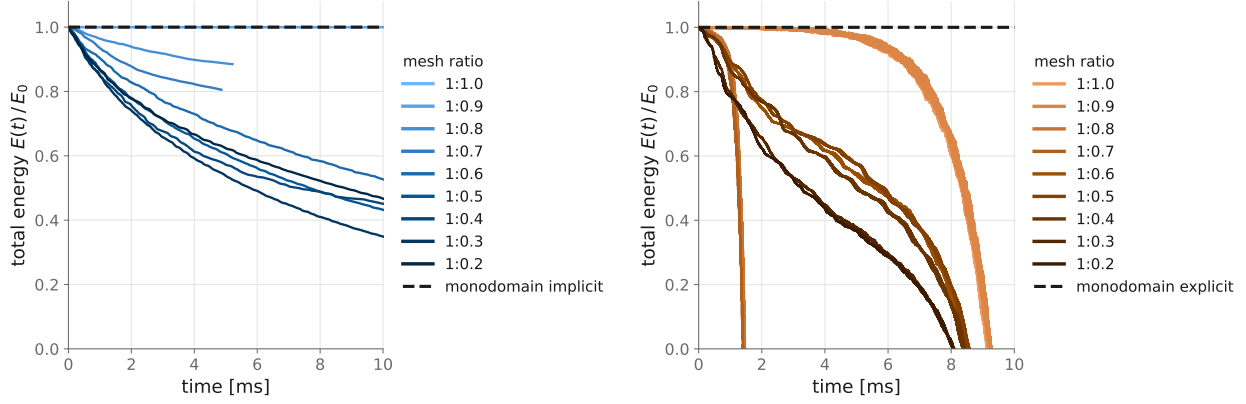


Figure 3: Adjoint-paired nonoverlap coupling, 10 ms, staggered metric. Left: implicit–implicit; no injection anywhere, conforming and 1:0.9 follow the monodomain reference, strong contrast dissipates monotonically, and 1:0.7 to 1:0.8 end at about 5 ms by element inversion. Right: explicit–explicit; the staggered kinetic term goes negative at every ratio as undamped grid-frequency content accumulates, and 1:0.7 to 1:0.8 fail by central-difference instability at 2.6 to 2.8 ms.

6.5 Windowed controller stops

With the subdomain steps fixed at $1\mu\text{s}$, controller stops of 1, 2, 5, and $10\mu\text{s}$ were compared to 10 ms. For the overlap family every trajectory is identical across the four windows (its exchange has no relaxation state). For the nonoverlap coupling the pre-fix per-pair relaxation state shifted the fixed point (Sec. 5.4); with the slot-keyed state every windowed policy reproduces the same-step trajectory to Schwarz truncation (a displacement mismatch of 3.9×10^{-6} at working tolerances, 3.7×10^{-8} at tightened ones; regression in test 110). Inside a window the policies were measured on the conforming II case at the $10\mu\text{s}$ window, in mean Schwarz iterations per stop: $\theta = 1$ needs 19.1 and completes; per-slot Aitken about 25 and fails at 3.9 ms; fixed $\theta = 0.5$ 47.5; Aitken pooled over the window’s slots 55.1 on II and fails on EE. The total number of substep solves is proportional to the iterations per stop whatever the window, and all windowed counts exceed the same-step Aitken count, so the controller step is a convenience for the nonoverlap coupling (for aligning output stops) and never reduces work; for the overlap family larger stops reduce it slightly. The explicit–explicit pair at large windows is impractical under any policy (the windowed iteration is nearly non-contractive and reaches the iteration cap).

6.6 The five couplings with the current code

The five couplings (overlap Dirichlet, nonoverlap Dirichlet–Neumann, overlap impedance, paired nonoverlap impedance, and the monolithic reference) were rerun to 10 ms with the code of 2026-09-17 at ratios 1:0.9, 1:0.75, and 1:0.5 with overlap 25.4 mm (Table 6, Fig. 4). The Dirichlet–Neumann exchange, not part of the earlier study, ends by element inversion within 3 ms with implicit subdomains at every ratio and conserves within 1% to 10 ms with explicit ones; why the same exchange is unstable with implicit subdomains and stable with explicit ones has not been explained (Sec. 8.3). Every other entry reproduces the corresponding earlier result.

Table 6: Cantilever, overlap 25.4 mm, code of 2026-09-17: $E(10 \text{ ms})/E(0)$, or the time the run ended by element inversion. “Bounded, noisy” is the explicit state of Sec. 1.4.

coupling	1:0.9 II	1:0.9 EE	1:0.75 II	1:0.75 EE	1:0.5 II	1:0.5 EE
monolithic	1.000	1.000	1.000	1.000	1.000	1.000
overlap, Dirichlet	3.16	fails 4.9 ms	fails 1.5 ms	fails 2.1 ms	fails 0.9 ms	fails 1.3 ms
nonoverlap, Dirichlet–Neumann	fails 2.9 ms	1.006	fails 0.4 ms	1.002	fails 0.7 ms	1.008
overlap, impedance	fails 7.2 ms	fails 8.4 ms	fails 6.0 ms	12.2	fails 7.3 ms	5.9
nonoverlap, impedance (paired)	1.000	bounded, noisy	fails 5.0 ms	fails 2.6 ms	0.43	bounded, noisy

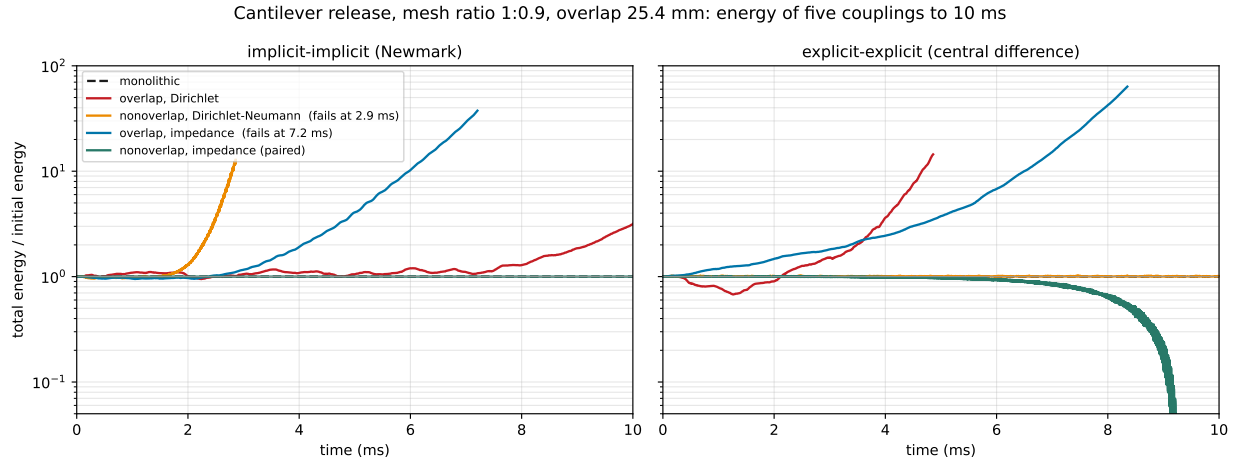


Figure 4: Cantilever at mesh ratio 1:0.9, overlap 25.4 mm, five couplings to 10 ms, code of 2026-09-17. Left: implicit–implicit. Right: explicit–explicit; the paired impedance curve ends in the bounded, noisy state.

6.7 The classical Robin–Robin condition

The Robin–Robin condition was run on both decompositions on 2026-09-18: **Schwarz RR nonoverlap** with the classical per-side transfer, and **Schwarz impedance overlap** with **impedance scale: 0**. Since the converged solution and the iteration both depend on α on nonconforming meshes, two values, one near the Dirichlet limit and one near the Neumann limit, were used: $\alpha = E/h = 10^{12} \text{ N/m}^3$ with $h = 6.35 \text{ mm}$, and $\alpha = E/L = 2.7 \times 10^{10} \text{ N/m}^3$ with $L = 254 \text{ mm}$. Integrators, ratios, relaxation, and tolerances are those of Table 6. Every run converged in 2 to 4 Schwarz iterations per stop, so Table 7 shows the converged coupling.

Table 7: Cantilever, classical Robin–Robin coupling, overlap 25.4 mm: $E(10 \text{ ms})/E(0)$, or the time the run ended by element inversion.

coupling	1:0.9 II	1:0.9 EE	1:0.75 II	1:0.75 EE	1:0.5 II	1:0.5 EE
overlap, $\alpha = E/h$	fails 0.7 ms	fails 1.4 ms	fails 0.8 ms	fails 1.9 ms	fails 0.4 ms	fails 0.8 ms
overlap, $\alpha = E/L$	fails 0.9 ms	fails 1.1 ms	fails 0.9 ms	fails 1.5 ms	fails 1.6 ms	fails 5.4 ms
nonoverlap, $\alpha = E/h$	fails 1.2 ms	fails 2.4 ms	fails 1.1 ms	fails 0.2 ms	fails 0.3 ms	fails 0.3 ms
nonoverlap, $\alpha = E/L$	fails 7.1 ms	0.96 (peak 1.05)	fails 7.4 ms	fails 4.5 ms	fails 3.4 ms	fails 0.9 ms

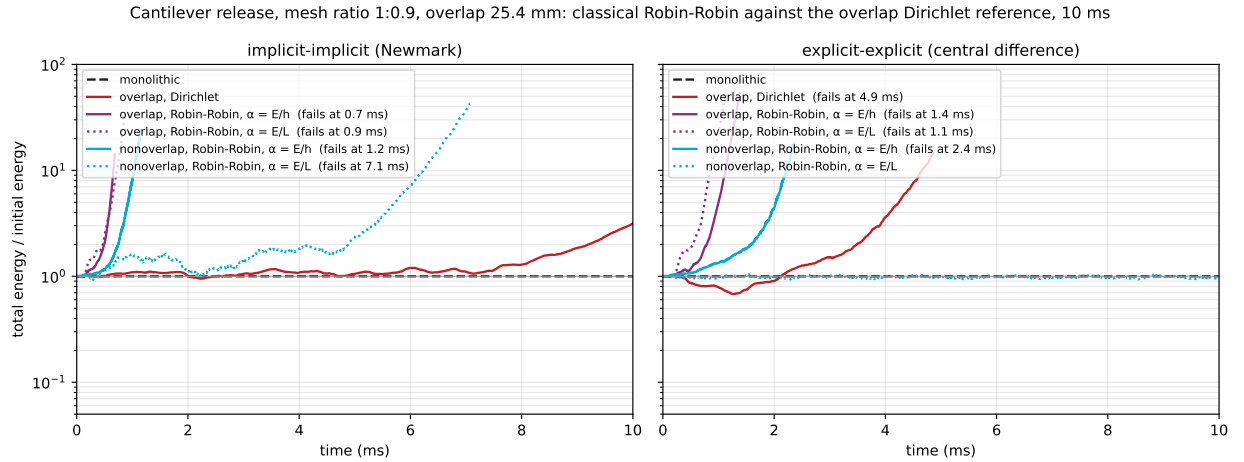


Figure 5: Cantilever at mesh ratio 1:0.9: classical Robin–Robin coupling on both decompositions at two Robin parameters, against the overlap Dirichlet reference, 10 ms.

Twenty-three of the 24 runs end by element inversion after exponential growth, between 0.2 and 7.4 ms; the one survivor (1:0.9, EE, $\alpha = E/L$) stays within 5% to 10 ms. The growth is faster than with overlap Dirichlet transmission in 18 of the 24 cases, and the stiffer spring fails sooner in 10 of the 12 pairs. The converged coupled solution depends on α (the two nonoverlap rows differ although both are converged to 10^{-8}): with per-side transfer on nonconforming meshes the discrete jump does not close, and the spring force on the residual jump scales with α . The result is the one (7) predicts: a condition that reflects every wave has no term that damps the interface mode fed by the discrete mismatch, and the spring stores that mismatch. The dashpot of the impedance condition is the term that removes it.

7 Measurements on the nested cylinders

The cylinders (Sec. 1.3) were built as a second case that excites the instability independently of the beam: three materials with a fiftyfold impedance contrast, a compressive front instead of flexure, and a closed curved interface with caps, at a mild mesh ratio (1:0.8). Every subdomain is explicit. The Schwarz iteration uses Aitken relaxation for the Dirichlet–Neumann exchange, a fixed factor $\theta = 0.5$ for the paired impedance and Robin–Robin couplings (Sec. 5.4), and no relaxation for the overlap couplings, with a relative tolerance of 10^{-6} and an absolute tolerance of 10^{-4} . The monolithic reference conserves the blended staggered energy to 0.01% over the 10^4 steps. Table 8 and Figs. 6–7 give the results.

Table 8: Nested cylinders, explicit on both sides, 5 ms: $E(t)/E(0)$ at the times shown, or the time the run ended by element inversion. Robin–Robin at $\alpha = E/h = 3.4 \times 10^{10}$ and $\alpha = E/L = 2.8 \times 10^9 \text{ N/m}^3$, with E the Young’s modulus of the filler, $h = 6.35 \text{ mm}$, and $L = 76 \text{ mm}$.

coupling	0.5 ms	1 ms	2 ms	5 ms
overlap, Dirichlet	1.02	1.29	fails 1.54 ms at $7.4 E(0)$	
overlap, impedance	1.02	1.76	fails 1.26 ms at $13 E(0)$	
overlap, Robin–Robin, $\alpha = E/h$	1.63	fails 0.66 ms at $5.5 E(0)$		
overlap, Robin–Robin, $\alpha = E/L$	1.53	fails 0.75 ms at $11 E(0)$		
nonoverlap, Dirichlet–Neumann	1.00	1.00	1.01	1.00
nonoverlap, Robin–Robin, $\alpha = E/h$	1.27	fails 0.81 ms at $3.1 E(0)$		
nonoverlap, Robin–Robin, $\alpha = E/L$	1.08	1.26	2.45	fails 3.59 ms at $11 E(0)$
nonoverlap, impedance (paired), 2 iterations per stop	0.79	0.54	0.24	0.065
same, 6 iterations per stop (to 1 ms)	0.89	0.77	—	—
same, $\theta = 1$	0.997 to 0.3 ms, then the iteration diverges			

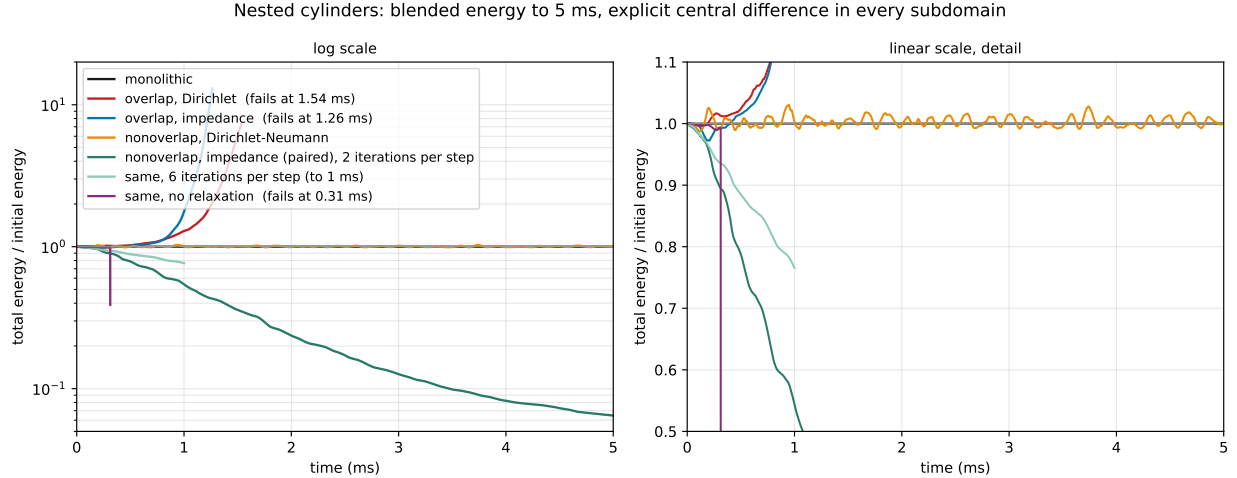


Figure 6: Nested cylinders, blended staggered energy to 5 ms: the Dirichlet and impedance couplings on both decompositions, with the two diagnostic variants of the paired coupling.

Dirichlet couplings. Overlap Dirichlet coupling leaves the reference after 0.4 ms, grows exponentially, and ends by element inversion at 1.54 ms at $7.4 E(0)$: the same mechanism and a comparable

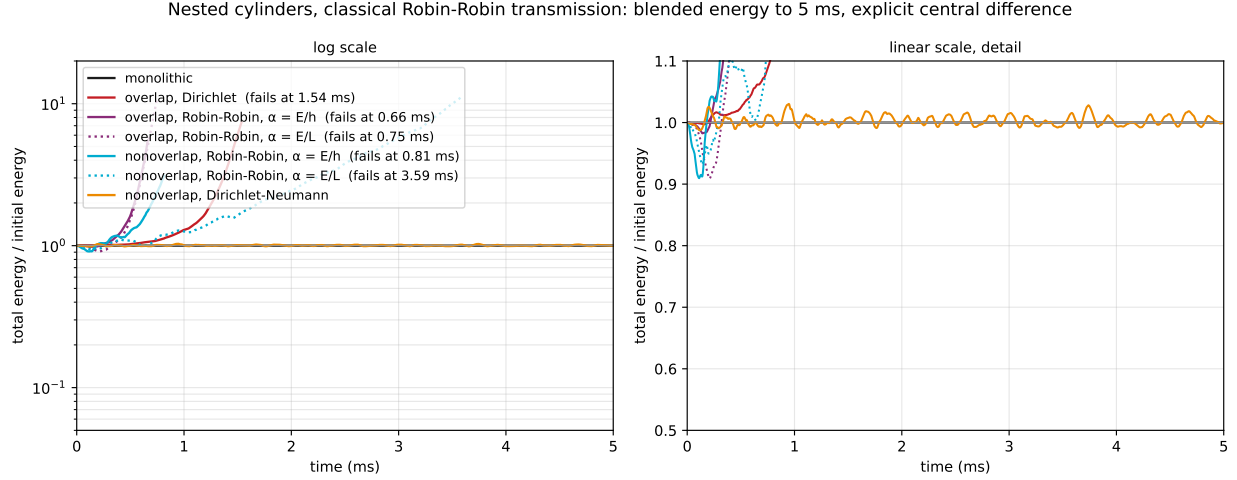


Figure 7: Nested cylinders, classical Robin–Robin coupling on both decompositions at two Robin parameters, against the overlap Dirichlet and nonoverlap Dirichlet–Neumann references.

rate as on the beam, although the interface here is closed and the meshes only mildly nonconforming. The nonoverlap Dirichlet–Neumann exchange (Aitken relaxation, 2 iterations per stop) stays within $+3$ and -1.3% for the whole 5 ms, as on the beam with explicit subdomains.

Impedance couplings. Overlap impedance coupling is unstable and faster than Dirichlet: it reaches $13 E(0)$ at 1.26 ms. On the beam the same condition survived 6 to 8 ms; here the dashpot faces the inter-grid discrepancy of a compressive front resolved with 5 and 6 mm elements. The paired nonoverlap coupling never grows but dissipates strongly: $0.54 E(0)$ at 1 ms and 0.065 at 5 ms. The interface velocity jump stalls at 0.1 to 0.14 of its initial size at every stop and is accepted by the displacement criterion, so the dashpot keeps absorbing it. Six iterations per stop keep $0.77 E(0)$ at 1 ms instead of 0.54, so about half of the loss is iteration lag on a jump the criterion does not measure; no relaxation keeps $0.997 E(0)$ to 0.3 ms and then fails to converge within 256 iterations. Every Aitken form diverges on this pair (Sec. 5.4).

Robin–Robin. All four runs end by element inversion after exponential growth. The overlap runs fail sooner than overlap Dirichlet (1.54 ms) and overlap impedance (1.26 ms); the nonoverlap runs fail where the paired impedance coupling never grows and the Dirichlet–Neumann exchange holds. The Schwarz iteration converged in 1 to 2 iterations per stop. As on the beam, the stiffer spring fails sooner, and the softer nonoverlap spring lasts longest, 3.6 ms, still growing to $11 E(0)$.

8 Conclusions

8.1 Findings

1. Dirichlet transmission on nonconforming meshes injects energy on both benchmarks, on the overlap decomposition and on the implicit nonoverlap Dirichlet–Neumann exchange: exponential growth ending in element inversion, at comparable rates on a flexural beam and on a shock-loaded nested assembly (Secs. 6.2, 6.6, 7).

2. The classical Robin–Robin condition is not a remedy. It reflects every wave ($|R| \equiv 1$), has no term that damps the interface mode, and pumps energy on both benchmarks and both decompositions, usually faster than Dirichlet transmission, at either Robin parameter; with per-side transfer its converged solution depends on α (Secs. 6.7, 7).
3. On an overlap boundary the impedance condition is consistent with the monodomain solution only with the partner traction, and with the consistent traction it is exact on conforming meshes (0.08% at 1 ms, no parameters). On nonconforming meshes its interface work is sign-indefinite in principle (two surfaces, non-adjoint transfers) and in practice (injection up to +275%, sign changes between adjacent ratios, 74 of 108 runs failing by 10 ms), and no construction at the interface changes that, because the discrepancy is the difference of the two grids’ dispersion relations, which both grids represent (Secs. 6.1–6.3). On the cylinders it fails sooner than Dirichlet coupling.
4. Energy consistency requires one interface traction acting through adjoint transfers built from one cross-mass matrix; the interface power is then the sign-definite dissipation (14) on the interface jump. Only the nonoverlap decomposition, with its shared interface, admits this construction (Sec. 4).
5. The adjoint-paired nonoverlap coupling never injects energy at any ratio, pairing, or benchmark. On the beam it conserves to 0.3% at 1 ms and to 0.0% at 10 ms on conforming and 1:0.9 meshes with an implicit member; at strong contrast it dissipates the unrepresentable jump at a persistent rate (40 to 85% by 10 ms); the non-nested ratios 1:0.7 to 1:0.8 accumulate interface-local high-frequency content until an element inverts at 3 to 5 ms; and explicit–explicit pairs end bounded but noisy, which subcycling the explicit member removes (Sec. 6.4).
6. On the cylinders the paired coupling is stable but loses half the energy within 1 ms, because the iteration stops on displacement while the interface velocity jump, which the dashpot damps, stalls at a finite value; more iterations halve the loss, and no relaxation nearly removes it before the iteration diverges (Sec. 7).
7. Different time steps per subdomain use piecewise-linear interpolation of the partner trajectory, whose dissipation of order Δt is the energy loss intrinsic to subcycling; exact cancellation of the subcycled interface work is incompatible with a partitioned exchange (Sec. 5.3).
8. The relaxation state must be keyed per pair and per substep slot; Aitken acceleration applies to single-slot stops only, reduces the iteration count tenfold on the implicit beam pairs, and diverges in every form on the explicit cylinders pair, where a fixed factor of 0.5 converges in 2 to 3 iterations (Sec. 5.4).
9. With explicit subdomains on both sides the plain Dirichlet–Neumann exchange conserved on both benchmarks (within 1% on the beam, 3% on the cylinders), while with implicit subdomains it is the fastest coupling to fail.

8.2 Recommendations

- **Energy-critical dynamic coupling:** use the adjoint-paired nonoverlap impedance coupling (the default of **Schwarz impedance nonoverlap**) with at least one implicit interface member and a conforming or nested mesh ratio; avoid non-nested mild ratios for long runs. Use a fixed relaxation factor with explicit pairs, Aitken with implicit ones.

- **Overlap decompositions:** conforming meshes with implicit–implicit integration conserve at any horizon; every other overlap configuration is for short transients only.
- **Robin–Robin:** for quasi-statics and comparison studies only.
- **All-explicit interfaces:** subcycle the finer side, or accept the bounded, noisy state; central difference removes no grid-frequency content at any stable step.
- **Parameters:** the impedance is material data (harmonic mean per pair for nonoverlap, partner material with P/S split for overlap); no scaling. The Robin parameter must be one value per interface under pairing and affects only the iteration when the jump can close; on non-conforming meshes its effect on the converged solution is unmeasured for the paired coupling, and the runs in this note use $\alpha = 0$ or $2 \times 10^9 \text{ N/m}^3$.
- **Diagnostics:** use the blended energy output with the staggered metric for explicit members, and the channel-resolved interface-work instrumentation as the acceptance metric for any change to the transmission machinery; a small net drift can be the difference of two large channels.

8.3 Open questions

- The stalled interface jump on the cylinders: a stopping criterion that refuses a stalled velocity jump, relaxation factors between 0.5 and 1, and subcycling of the explicit side are the candidates for recovering the energy the paired coupling now dissipates.
- The dependence of the converged paired solution on α and on θ : the fixed point differs by about 4×10^{-5} relative displacement between $\theta = 0.5$ and 1 at a 10^{-13} tolerance on the 1:0.5 beam, evidence of a neutral mode of the exchange that the displacement criterion does not measure; an α sweep of the paired coupling on nonconforming meshes has not been run.
- Why the Dirichlet–Neumann exchange is stable between explicit subdomains and unstable between implicit ones.
- The non-nested mesh-ratio failure window of the paired transfer at long horizons.
- Exact subcycled interface-work cancellation through a direct interface solve, with the localized Lagrange multiplier frame construction [12, 13].
- Mesh-ratio and overlap-width families on the cylinders, and implicit versions of its subdomains, to test whether the beam conclusions transfer.

Data provenance

Correction-sequence study and instrumentation: `rigel:~/test/Norma/cantilever-energy`. Mesh-ratio sweep, 10 ms horizon, controller-step sweep, and run harness: `sirius:~/cantilever-energy-study/`. The 2026-09 reruns, the Robin–Robin runs, and the cylinders runs: `rigel:~/test/Norma/beam/ (runs, runs-r090, runs-r050)` and `rigel:~/test/Norma/cylinders/`. Slides: `Norma-Energy-Stability-Beam-Cylinders-2026-09.pptx`. Regression tests: 104 (overlap impedance energy and consistent-traction patch), 105 (tensor impedance cross-scaling), 106 (variational transfer projectors and content-absorption dissipativity), 110 (adjoint pairing, windowed equivalence, keyword guards), 125 (explicit stable step with subcycling).

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